



Python for Fixed Income

COURSE 4 · MACHINE LEARNING

Hierarchical Clustering and Cluster Profiling

WEEK 3 · DAY 3

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From the top down to the bottom up

Yesterday: K-means

Choose k first. Four centres move until every one of 821 bonds sits with its nearest centre. Four labels come back.

Today: hierarchical, on issuers

150 issuers, each a cluster of one. Join the two closest, repeat until one is left. Every merge is kept: choose k by cutting the tree.

Then the profile: counts and medians in bp, years, and billions, the part a desk reads.

Invented issuers and generated spreads: a cluster is a group of similar rows, never a ranking.

One row per issuer, and a distance by hand

Column	From the bonds
median_log_oas	median of log OAS
median_duration	median, years
bonds	how many bonds
amount_bn	sum outstanding, USD bn
score	issuers.csv rating, 1 to 18

```
distance = sqrt(sum of  
(difference in each scaled column)^2)
```

2.0227

BEZE to BEZH

0.2075

closest: CLAY and WYXR

11,175 pairs of issuers. The closest pair is the first merge.

Scaled first: a difference of 1 is one standard deviation in any column.

Merging step by step

Merge	Joins	With	Height
0	CLAY	WYXR	0.2075
1	ELVX	GALY	0.2586
2	BEZN	VYRN	0.3289
3	THAX	XANH	0.3628
4	HESA	KEZA	0.4167
5	CLAL	YARN	0.4331
6	FESR	NEVN	0.4447
7	CLAX	ELVN	0.4491

How scipy numbers clusters

0 to 149: single issuers. The cluster made by merge i is $150 + i$. Two single issuers join at their distance.

Merge 42: the first with a formed cluster

RULE joins cluster 169 at 0.7167, making 3. Every merge before it pairs two single issuers.

Linkages: four meanings of close

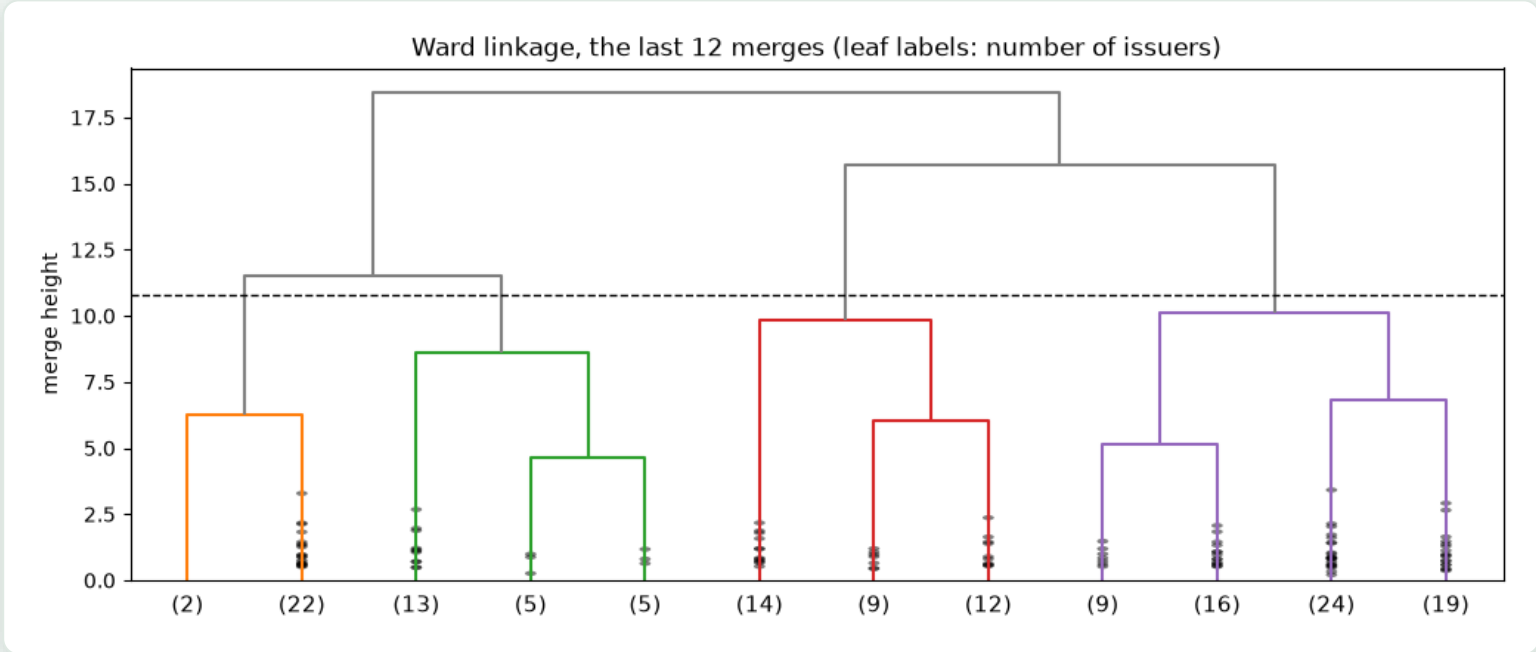
Linkage	Distance between two clusters
single	the closest pair
complete	the farthest pair
average	the average over all pairs
ward	the rise in squared distance to the centre

Cut at 4	Sizes, largest first
single	145, 2, 2, 1
complete	64, 35, 28, 23
average	115, 30, 3, 2
ward	68, 35, 24, 23

Single linkage chains: 145 of 150 issuers in one cluster. Ward gives groups of even size.

Same distances, four rules, four trees. The course uses ward, Euclidean, on scaled columns.

The dendrogram, and where to cut



Height	Cut below: k
18.4445	2
15.6804	3
11.4930	4
10.1186	5

Cut at 10.81: 4 clusters

A judgement. The widest gap says 3; the silhouette says 2. Four keeps day 12's k and splits by duration.

Cophenetic correlation

Cophenetic distance: the height at which the tree first joins two issuers. Its correlation with the real distances, over every pair, says how faithfully the tree keeps them.

Linkage	Cophenetic
average	0.6317
ward	0.5212
complete	0.4903
single	0.4637

Two pairs

CLAY and WYXR: distance 0.2075, cophenetic 0.2075. The first merge.

BEZE and BEZH: distance 2.0227, cophenetic 15.6804. Different branches.

Average keeps distances best, yet its cut gives 115, 30, 3, 2. Look at the cut too.

Cutting the tree with fcluster

```
fcluster(merges, t=4, criterion="maxclust")      # the course's convention  
fcluster(merges, t=cut_height, criterion="distance") # the same clusters here
```

68

cluster 0

35

cluster 1

24

cluster 2

23

cluster 3

Numbered by size, largest first; a tie goes to the cluster holding the earlier row. AgglomerativeClustering, ward, 4: adjusted Rand index 1.0000, the same groups.

Cluster 0 is first in size, and in nothing else.

The profile, in the data's own units

Cluster	Issuers	OAS (bp)	Duration (yr)	Bonds	Amount (\$bn)	Score
0	68	124.5	6.48	7.0	8.70	9.0
1	35	342.0	5.23	4.0	6.05	14.0
2	24	68.0	7.32	3.5	4.03	5.0
3	23	73.0	3.57	5.0	7.50	6.0

cluster_profile: count and medians of the unscaled rows. 53 tests pass. Clusters 2 and 3: similar spreads, split by duration.

A profile describes a group. It never ranks one.

K-means against hierarchical, same issuers

Hierarchical	K 0	K 1	K 2	K 3
0	36	12	5	15
1	0	20	1	14
2	0	1	23	0
3	9	5	9	0

0.2552

adjusted Rand
index

0.2153

silhouette, ward

0.2577

silhouette, K-means

Read across: ward's cluster 2 lands almost whole in one K-means cluster; ward's wide-spread cluster 1 is split by size.

Neither set of groups is the true one: two rules impose four groups on these issuers.

What a profile may say

Allowed names

cluster, cluster 2

a group of similar rows

counts and medians in units

Never called

good, bad, safe, risky

cheap, rich, mispriced

a ranking, a signal, a trade

Median log OAS and score: correlation 0.97, credit counted twice. Bonds and amount: 0.82, both random draws in the generator.

A group is only as meaningful as the columns that built it.

Excel side by side

Excel side by side (tested in Excel)

`=STANDARDIZE(B2,AVERAGE(B$2:B$151),STDEV.P(B$2:B$151))`

`=SQRT(SUMXMY2(J2:N2,J3:N3))`

`=SQRT(SUMXMY2(J18:N18,J137:N137))`

`=COUNTIF($H$2:$H$151,0)`

Pivot table: Count, Average

`=MEDIAN(IF($H$2:$H$151=0,G$2:G$151))`

equal to StandardScaler

2.0227, BEZE to BEZH

0.2075, CLAY to WYXR

sizes 68, 35, 24, 23

no median in its list

median OAS 124.5 bp, exact

Excel has no hierarchical clustering: the tree is built in Python, the numbers around it in Excel.

Save the day: the size of a commit

1. Module

mlkit.py, test_mlkit.py, .gitignore

2. Results

results.csv and profile.csv, 4 decimals

3. Stat

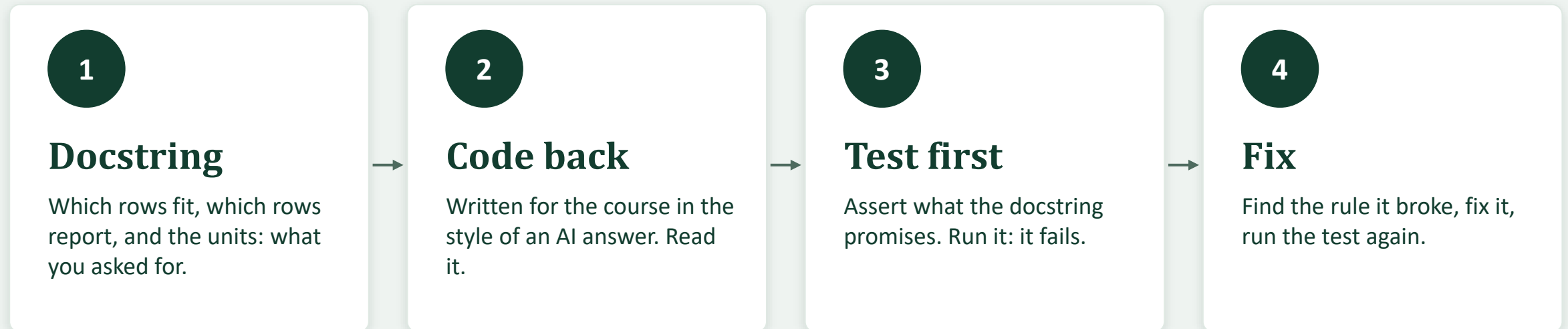
git diff HEAD~1 --stat

```
profile.csv | 5 +++++  
results.csv | 2 ++  
2 files changed, 7 insertions(+)
```

HEAD~1 is the commit before the last; --stat is one line per file. Read it before a long diff.

The AI check

The docstring is the prompt. The code that comes back runs, and has one modelling bug.



Today: the profile is computed on the scaled columns and labelled median OAS in bp: 1.3502 for a cluster at 342.0 bp.

Code that runs is not the same as code that is right.

What you can do now

- Start with every issuer alone; merge the closest, 149 times
- The linkage decides what close means for a group
- Cut where you judge best, with reasons; number by size
- Cophenetic correlation: the tree, not the cut
- Profile in bp, years, billions; never rank a cluster

Week 3, Day 4 PCA and t-SNE: summarise the curve's columns, picture the bonds